ZCAS University

CIT5241 INFORMATION RETRIEVAL AND WEB SEARCH

MID TERM TEST

22ND APRIL 2026

08 00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Answer any **TWO** questions from the three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

Question 1 (25 Marks)

Scenario: Job applicant resume screening system.

a) Describe with the aid of an example the procedure employed in the vector space model to generate rankings of documents given a particular query. Use job recruitment terms (e.g., "Python", "project management", "data analysis") in your example. **[5 Marks]**

b) Using your answer in (a), generate a ranked list given the query and the list of documents below. Show all working (term frequency, inverse document frequency, cosine similarity, etc.).

Query: "software engineer with Python and SQL"

Document 1: "Python developer with database experience in SQL and NoSQL"

Document 2: "Frontend engineer specializing in JavaScript and React"

Document 3: "Data engineer using Python, SQL, and cloud platforms"

Document 4: "Software engineer experienced in Java and C++ only"

Document 5: "Full-stack developer proficient in Python, SQL, and Django"

[20 Marks]

Question 2 (25 Marks)

Scenario: Customer review database for an e-commerce platform.

Consider these documents:

Document 1: "battery life is excellent but screen is fragile"

Document 2: "poor battery life and slow charging"

Document 3: "excellent screen quality and fast charging"

Document 4: "slow charging and fragile screen, but battery is good"

a) Draw the term-document incidence matrix for this document collection. **[10 Marks]**

b) Draw the inverted index representation for this collection. **[10 Marks]**

c) What is the returned result for the following query below? **[5 Marks]**

Query: "battery AND excellent NOT fragile"

Question 3 (25 Marks)

The Boolean Retrieval Model is a foundational approach in Information Retrieval (IR) that uses Boolean logic to retrieve documents based on exact match criteria.

i. Explain the concept of the Boolean Retrieval Model in Information Retrieval. How does it differ from other retrieval models like the Vector Space Model? **[5 Marks]**
ii. Describe the key components of the Boolean Retrieval Model, including terms, documents, and Boolean operators. Provide examples to illustrate each component. [7 **Marks]**
iii. Discuss the three main Boolean operators used in the Boolean Retrieval Model: AND, OR, and NOT. Explain how each operator influences the retrieval process. **[6 Marks]**
iv. Provide an example of a Boolean query and step-by-step describe how the query is processed to retrieve relevant documents. Use at least two Boolean operators in your example**. [7 Marks]**

(25 marks)

3
S
C
-
V
L